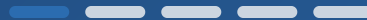




AI Spec Development

Week 8 • Day 1 • AI Spec Development & Capstone

TPM Academy — the final week



Day 1 Objectives

- Internalize the **AI Spec Development pattern** — a 5-prompt sequence with validation
- Walk through the **AI Spec template** (8 sections) and map each to inputs from prior weeks
- **Practice** the sequence on FieldPulse where inputs are validated
- Confirm the **capstone subject** through the 5-question fit test
- Plan the **discovery work** needed before tomorrow's compressed PRD



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin the 5-artifact set + capstone candidates
09:15 – 10:30	Block 1 + Activity 1	Capstone subject confirmation
10:45 – 12:00	Block 2 + Activity 2	Walk through AI Spec template
13:00 – 14:15	Block 3 + Activity 3	Run 5-prompt sequence on FieldPulse
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Capstone discovery plan

Block 1 — Capstone Subject Confirmation

5 questions; refine or commit

The 5 Capstone-Fit Questions

1. Do we have a **real customer**? (interviews / observation / data)
2. Is the scope **substantive enough**?
3. Can you state the **core problem** in one sentence an engineer could scope from?
4. Can we **discuss it publicly**?
5. Can we **ship the artifact set in 4 days**?

If 1+ is "no": refine or pick a backup. A subject that fails this test produces a thin capstone.

Activity 1 — Subject Confirmation

Triads • 35 minutes

- 5 min: re-read declared candidate
- 15 min: 5-question fit test
- 10 min: adjust or commit
- 5 min: list inputs needed



5 + 15 + 10 + 5

Block 2 — The AI Spec Pattern

What it is, what it isn't, what's in the template

AI Spec Development

Is	Is not
Structured sequence of validated prompts	"Just paste everything and summarize"
Anchored on PRD/TCD/TMD/SEP/DP outputs	Replacing the artifact set
Disciplined: every output validated	Trust the AI's output
Iterative: 4–6 prompts in sequence	One mega-prompt
Engineering-ready output	A pretty draft for stakeholders
Always paired with provenance log	Untracked AI use

Senior TPMs use this to draft, not to think. Judgment shapes every section; AI accelerates the typing.



AI Spec Template (8 sections)

1. **Headline** — one sentence
2. **Engineering-ready summary** — 3–5 paragraphs
3. **Data + API contract**
4. **Sequence + failure handling**
5. **Constraints** (perf / sec / compliance / a11y)
6. **Decisions made (and not made)**
7. **Stakeholders + sign-off**
8. **Provenance log**

Shorter than the sum of its inputs. The synthesis with citations to the deeper artifacts.

Activity 2 — Walk the Template

Triads • 40 minutes

- 5 min: read template
- 30 min: walk each section, map to input artifacts
- 5 min: identify "section we'll struggle with" + counter-measure



5 + 30 + 5

Block 3 — The 5-Prompt Sequence

Each prompt has a job + validation step

The Sequence

#	Prompt job	Inputs
1	Headline + summary	PRD §§1–5, TCD §1, TMD §3
2	Data + API synthesis	TMD §1 + TMD §3
3	Sequence + failure	TMD §4
4	Constraints	TCD §3 + TCD §4 + PRD §7
5	Decisions + stakeholders	TCD §5 + TCD §6 + SEP §1

Each prompt validates before the next runs. The validation chain is what keeps the AI Spec honest.

Validation Discipline (cumulative from Wk 5 D5)

- **Cross-checked all citations** — best
- **Spot-checked** — rougher
- **Adopted with rationale** — correctness self-evident
- **Pending** — track to closure
- **Rejected** — cut

Skip the validation step and you ship hallucinated specs. Catastrophic in real engineering.

Activity 3 — Run on FieldPulse

Triads • 40 minutes

- 30 min: run 5 prompts in sequence; validate each output
- 10 min: assemble the AI Spec + provenance log



30 + 10

Block 4 — Capstone Discovery Plan

What you need to gather before tomorrow



4 Discovery Categories

1. **Customer signal** — interviews, support data, public reviews
2. **Strategic context** — why this matters, what's been tried
3. **System inputs** — existing docs, codebase access, related artifacts
4. **Stakeholder signal** — who has opinions; what are they

"We don't have this and can't get it" — document honestly. Mark for AI-augmented hypothesis tomorrow with extra validation.



Capstone Plan AI Check

Role: Capstone advisor for a TPM cohort.

Context: <capstone subject="" += "" 5-question="" fit="" assessment="" declared="" inputs="" needed="">

Task: Identify 3 risks in the capstone plan:

1. A customer signal source likely insufficient
2. A scope dimension likely too ambitious for 4 days
3. A stakeholder whose buy-in we'll need but didn't list

Constraints:

- Be specific to the subject given
- Suggest a concrete adjustment per risk

Format: 3 numbered findings.</capstone>

Activity 4 — Discovery Plan

Triads • 45 minutes • Wrap

- 10 min: list inputs needed (4 categories)
- 10 min: distribute work across triad
- 5 min: deadline for inputs gathered
- 10 min: identify "we don't have this" honestly
- 10 min: AI risk-check + adjustments



10 + 10 + 5 + 10 + 10 • 15-min wrap



Day 1 Wrap

What we built

- AI Spec template (8 sections)
- 5-prompt sequence with validation
- FieldPulse practice run
- Confirmed capstone subject
- Discovery plan

What opens tomorrow

- **Capstone discovery + compressed PRD**
- Customer interviews / signal
- PRD-light (sections 1–7)

Bring to Day 2: All discovery inputs gathered. Tomorrow we draft the capstone PRD.

Questions & Reflection

Which input does your capstone need that you don't yet have?